Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Examination, 2018

Hum 1117

(Functional English)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make sentences using the following Modals as directed. (14)
   - i. Could (To express polite request)
   - ii. Could (To express past ability)
   - iii. Must (To express obligation)
   - iv. Must (To express logical deduction in the past)
   - v. Should (To express duty in the past)
   - vi. Need (To express unnecessary action in the present)
   - vii. Had better (To express preference)

   (b) Make sentences on the following structures using the verb given in brackets. (12)
   - i. Subject + Intransitive verb +adverbial of manner. (Talk as verb)
   - ii. Subject+ Linking verb + adj. complement. (Turn as verb)
   - iii. Subject+ Linking verb + noun complement. (Turn as verb)
   - iv. Subject+ Transitive verb +Gerund as object. (Stop as verb)
   - v. Subject+ Transitive verb +Infinitive as object. (Want as verb)
   - vi. Subject+ Transitive verb + object+ adj. complement. (Declare as verb)

   (c) Define participle, gerund and infinitive. Give two examples of each of them in sentences. (09)

2. (a) Frame Wh questions from the underlined parts of the following sentences. (14)
   - i. Nobody believes a liar.
   - ii. Riches do not always bring peace.
   - iii. An ambulance carried away the patient to hospital.
   - iv. Because of cloudy weather, we postponed our plan.
   - v. I waited there for three hours.
   - vi. I read in class six.
   - vii. It is half past twelve now.

   (b) Transform the sentences as directed. (12)
   - i. Nobody had seen her smile. (Passive)
   - ii. He kept the promise he made. (Simple)
   - iii. Go there or you will be punished. (Complex)
   - iv. Of all the rivers, the Padma is the longest. (Positive)
   - v. Nobody was displeased with me. (Affirmative)
   - vi. If you speak the truth, I shall let you go. (Compound)

   (c) Make sentences using the following phrases and idioms. (09)
   At a stretch; Bad blood; Hard and fast; In a body; On behalf of; Ad hoc.

3. (a) Make sentences using the following words as directed. (14)
   Last (as noun); Last (as adjective); Last (as verb); Last (as adverb); Down (as adjective); Down (as adverb); But (as adverb).

   (b) Change the following words as directed and make sentences with the changed words. (12)
   Danger (into verb); Enemy (into noun); Face (into adjective); Gain (into noun); Gain (into adverb); Idiot (into adjective).

   (c) Supply a suitable word to fill in the blanks. (09)
   - i) He prides .... on his success.
   - ii) He specialises .... psychology.
   - iii) .... do you think the boys love most?
   - iv) Their dreams have been ....
   - v) What .... do you want?
   - vi) They are .... for a man.

4. (a) Correct the following sentences. (14)

i. Chairman is attending at the meeting.
ii. The physics is interesting to study.
iii. Are you his elder?
iv. Rana will go to Khulna, Jessure, Magura and others.
v. You what say is fine.
vi. He not only ate a banana but also a cake.
vii. He is behaving friendly with all in a society.

(b) Express the following notions/function in sentence. (12)

i. Love; ii. Sympathy; iii. Duty; iv. Wish; v. Congratulations; vi. Pride

(c) Define transitive verb, intransitive verb and linking verb with two examples following each of the definitions. (09)

Section – B

5. (a) Read the following passage and answer the questions that follow: (20)

A young lad was passing along a lane in Banaras. He had some fruits in his hands. A number of monkeys followed him. Soon they became so bold that they tried to snatch away the fruits. The lad began to run in fear. This only made the monkeys bolder still. They ran after the boy and were about to fall upon him. Just then the lad heard a voice, "Don't run my boy. This only puts you in greater danger. Face the monkeys, and they will stop." Looking up, the lad saw a Sannyasi close by. He took heart and turned round to face the monkeys. They stopped at once. He moved a few steps towards them, and they fell back. He then walked on with bold steps through the monkeys. They fell back still more, and they went away. The lesson went deep into the lad's mind. In after times he became famous as Swami Vivekananda. He would then speak of it very often. He taught: "Be not afraid of anything. The moment you fear you are gone. It is fear that is the cause of misery in the world."

Questions:

i. Who was the young lad? Why did he run?
ii. What was the result of his running?
iii. What did the lad do after hearing the voice and what was the result?
iv. What was the lesson he learnt?

(b) Make a précis of the above passage (Q5.a) with a suitable title. (15)

6. (a) Amplify the idea "Patriotism in not enough." (20)

(b) Write a paragraph on industriousness. (15)

7. (a) Prepare a report on Medical Centre of KUET. (20)

(b) Write a letter to your younger brother on the merits of Facebook. (15)

8. Write a free composition on the following: (35)

(a) Quality education and the future of a country

or

(b) Social awareness and development

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BURP 1st Year 1st Term Regular Examination, 2018

Math1117
(Mathematics - I)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define the following matrices with an example for each: (09)
   (i) Idempotent matrix
   (ii) Orthogonal matrix
   (iii) Skew-Symmetric matrix

   (b) Find the symmetric and skew-symmetric parts of a matrix, $A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{bmatrix}$ (11)

   (c) Show that $A\,.(adj\,A) = A\,.\,I_n$. If A is not singular matrix, then find the value of $|A\,.\,adj\,A|$. (10)

   (d) Define rank of a matrix with example. (05)

2. (a) Reduce the following matrix to its echelon form, canonical form and finally normal form. $A = \begin{bmatrix} 0 & -4 & 1 & -2 \\ 1 & 2 & -1 & 1 \\ 2 & 0 & -1 & 0 \end{bmatrix}$ (20)

   If A be an augmented matrix of a system of linear equations, then, if possible, find the solution of the system. Also for initial augmented matrix A, express it as a system of linear equation in algebraic form.

   (b) Define non-singular matrix. Find the inverse of a matrix, $A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$. (15)

   Let $AX = H$, where $H = [1 \quad 0 \quad 1]^t$, then find the solution for x.

3. (a) If possible find the value of m so that the following system of linear equations has (i) no solution (ii) unique solution (iii) many (arbitrary) solutions. Give $2mx + y = m; 2x + 3my = 1$. In the case of existing solution for some m, find the solution(s). (15)

   (b) You have a circular region which passes through the fixed point: A (4, -3), B (-2, 7) and C (-4, 5) for an urban plan. Formulate the problem as a system of linear equation and hence solve it by using matrix operation. Moreover using the concept of geometry, find the centre of the circle. (20)

4. (a) Draw e^x and e^{-x} on a graph paper by considering some values of x. Hence draw $\sinh x$ and $\cosh x$ on the same graph for corresponding (20)

values of x. Finally sketch $\tanh x$ on the same graph paper.

(b) Define logistic function/curve (including meaning of the parameters). Show the figures of logistic curve for different numerical values of steepness parameter k. Also establish the relation between standard logistic function and hyperbolic tangent i. e. $2f(x) = 1 + \tanh\left(\frac{x}{2}\right)$ (15)

Section – B

5. (a) Find the transformed equation of $4xy - 3y^2 - 9 = 0$ if the axes are rotated through an angle $\tan^{-1}(2)$. (11)

(b) Identify the conic $2x^2 + y^2 - 3xy - 5x + 4y + 6 = 0$. Also find its centre if it is a central conic. (12)

(c) Show that the equation $2x^2 - y^2 + 2z^2 - yz + 5zx + xy = 0$ represents a pair of plane. Also find the angle between them. (12)

6. (a) Find the equations of plane which passes through the line of intersection of two planes $x + 2y + 2z = 1$ & $x + y - z + 1 = 0$ and is at a unity distance from the point (4, -2, 1). (12)

(b) Find the coordinates of the point where the line joining the points (2, -3, 1) and (3, -4, -5) cuts the plane $2x + 3y + z = 10$. (12)

(c) Define direction cosines of a line. Find the direction cosines of AP and AQ, where the coordinates of A, P, Q are A (2, -3, 4), P (7, 3, 2) and Q (3, 2, 8). Also find the angle between AP & AQ. (11)

7. (a) Find the perpendicular distance from the point (1, -2, 3) to the line which passes through the point (2, -3, 5) and makes equal angles with the axes. (11)

(b) Find the symmetrical/standard form of the equation of a line $5x + 2y - 2z + 1 = 0 = x - 2y - 6z + 17$ and also find its direction cosines. (12)

(c) Determine the length of the shortest distance (S. D) between two skew-lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$. (12)

8. (a) Define spherical triangle. For any spherical triangle prove that $\cos a = \cos b \cos c + \sin b \sin c \cos A$. Transfer the formula for spherical triangle. (13)

(b) Solve the spherical right angled triangle ABC, when $< C = 90°$ $a = 119°46'36''$ and $< B = 52°25'38''$. (11)

(c) Find the great circle distance between the two place $A\ (21°20'N, 89°24'E)$ and $B\ (30°20'N, 38°47'E)$. (11)

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BURP 1st Year 1st Term Examination, 2018

Ch 1117

(Environmental Chemistry)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable value for any missing data.

Section – A

1. (a) What is shielding effect? Discuss the factors that affect the magnitude of shielding effect. (08)

(b) Discuss the anomalous position of Be with the alkaline earth metals. In what respects Be is different from other alkaline earth metals and why? (08)

(c) Indicate the different types of bonds that are present in the following compounds: (09)

(i) $NaCl$, (ii) $(H_2O)_x$ (iii) $[Cu(NH_3)_4Cl_2]$

(d) Write short notes on: (10)

(i) Isomorphism (ii) Resonance structure

2. (a) What is nuclear binding energy? Explain why the intermediate nuclei are stable, where as the lighter nuclei and heavy nuclei are unstable. (10)

(b) Define mass defect. Explain with diagram, how nuclear reactor works? (10)

(c) If a watch contains a radioactive substance with a decay rate of $1.4 \times 10^{-2}\ y^{-1}$ and after 50 years only 25 mg remains, calculate the amount originally present. (07)

(d) Write the applications of radio-chemistry in agricultural and industrial sectors. (08)

3. (a) What is solution? Write different types of solutions with examples. (09)

(b) What is critical solution temperature? Explain the effect of temperature on the mutual solubility of Phenol-water system, with appropriate diagram. (10)

(c) 5g of NaCl is dissolved in 1000g of water. If the density of the resulting solution is 0.997g per ml, calculate the molality, molarity, normality and mole fraction of the solute, assuming volume of the solution is equal to that of solvent. (10)

(d) Explain "solubility of non-ionic (sugar) substances in water". (06)

4. (a) Draw the molecular orbital diagram of O_2^+ and calculate its bond order. (09)

(b) Classify the following substances as acids or gases, giving reasons: (10)

Ag^+, SO_3, KCN, OH^-, Cl_2

(c) Why Li is the best reducing agent than Cs? (09)

(d) Explain why alkali metals are highly reactive? (07)

Section – B

5. (a) 3° carbocation is more stable than 1° carbocation. Why? (07)
   (b) Discuss the stereo chemistry of S_N1 reaction mechanism. (10)
   (c) An S_N2 reaction proceeds with complete stereo chemical inversion- explain. (10)
   (d) What do you mean by the terms: (08)
   (i) Substrate (ii) Attacking reagent and (iii) Intermediates
   Apply these terms to a substitution reaction.

6. (a) Discuss the free-radical mechanism of polymerization process. (10)
   (b) What is Zieglar-Natta Catalyst? Discuss its application in terms of "stereo chemical control". (10)
   (c) Discuss with example the chain reaction polymerization and step reaction polymerization. (10)
   (d) Write short note on 'Co-polymerization'. (05)

7. (a) Write down the characteristic properties of oligotrophic lake and eutrophic lake. (10)
   (b) Discuss the domestic waste water treatment process. (12)
   (c) How would you broadly divide the major regions of the atmosphere? State their respective altitudes and temperature ranges. What are the important chemical species in each region? (07)
   (d) What is environmental chemistry? Write down the composition of atmosphere. (06)

8. (a) Discuss the electro-chemical mechanism of wet corrosion. (10)
   (b) Describe in brief the prevention of corrosion by the modification of the environment. (11)
   (c) With help of a diagram describe the effect of different factors on atmospheric corrosion. (08)
   (d) Define the followings: (06)
   (i) Pollutant (ii) COD and BOD (iii) Industrial hazards

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BURP 1st Year 1st Term Regular Examination, 2018.

URP 1113

(Fundamentals of Planning Process)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Explain the concept of "Planning". "Planning is more needed in a developing country like Bangladesh. "– Explain. (10)
   (b) "Planners' role is changing in different ways from the beginning of planning till now." Provide a brief overview of that scenario. (15)
   (c) Define "Paradigm Shift" in planning theory. Explain with appropriate example. (10)

2. (a) Explain the concept of Mixed Scanning Approach of Planning with an appropriate example. (10)
   (b) Write down the comparison of steps and principles between Rational and Incremental Planning. (10)
   (c) Recall one episode in your life in which you feel you have made a decision rationally. Describe how you arrived at your decision and fit your decision and fit your decision making process with the theoretical model of rational decision making process. (Hint: Choosing a Major , A place of live, A Brand of Bicycle to Buy) (15)

3. (a) Illustrate the Advocacy Planning process with the help of a case study. (15)
   (b) Write down the role distribution (of planners and of population) in a Transactive Planning process. (10)
   (c) Explain the characteristics of Radical Planning Theory. (10)

4. Write short note on any 5 (five) of the following : (35)
   a) Top - down and Bottom - up approach
   b) Criticisms of Comprehensive Planning
   c) Radical Planning
   d) Negotiator
   e) Factory towns
   f) Limitations and strengths of Advocacy Planning
   g) Components of Transactive Planning

Section – B

5. Annotate any five of the following : (35)

   a) Vision and Mission
   b) Goal and Objectives
   c) Urban development plan
   d) Project
   e) Programme
   f) Strategic Plan
   g) Detailed Area Plan

6. (a) "Planning is related to three key aspects (social, economical and physical environment)" – briefly explain the statement with proper example. (10)

   (b) "Planning has three dimensions in terms of Space, Sector and Time". Briefly explain with relevant examples in the context of Bangladesh. (15)

   (c) What is understood by Urban Area? What are the conditions to be an urban area (Municipality) as per as the Pourashava Ordinance 2009? (10)

7. (a) The lifecycle of urban development contains several phases .Briefly explain the phases with necessary examples. (20)

   (b) Differentiate between Structure Plan and Master Plan with appropriate examples. (15)

8. (a) Draw the figure "A Ladder of Citizen Participation" and explain the stages with relevant examples. (20)

   (b) "Plan (in context of urban and regional planning) should be Specific, Measurable, Achievable, Realistic and Time Bound ". – Why? (15)

Khulna University of Engineering & Technology
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BURP 1st Year 1st Term Examination, 2018
URP 1111
(History/Evolution of Human Settlement)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define human settlement. Mention the major innovations of the Stone Age. (15)
   (b) What led Man to change his diet and eat grains and small animals? (05)
   (c) Write short notes on the following: (15)
   (i) Metallurgy (ii) Solomonic Fortifications (iii) Sedentary life style.
2. (a) Distinguish between urban and rural settlement. (10)
   (b) Classify towns based on their functional classification and rural settlements on pattern. (15)
   (c) Explain the reasoning of ancient people from the following statements- (10)
   i) People developed their settlements along sea coasts and mountains.
   ii) Circular pattern of the roads at the center of the settlement.
3. (a) Draw a "pizza slice" of the Garden City with description. (15)
   (b) Briefly discuss the concept of the Three Magnet Theory. (05)
   (c) Describe the Vertical City of Le Corbusier. Also write about the Street system in Plan Voisin. (15)
4. (a) Briefly elaborate about the linear city concept and theories by Arturio Soria. (10)
   (b) "The growth and development of human settlement depended on many reasons in ancient Bengal"-find out the reasons. (10)
   (c) Write short notes on: (15)
   (i) Pundravarddhan (ii) Samatata (iii) Vanga

Section – B

5. (a) Write down some names of unique human settlements. Why are they unique? Describe your point of view. (08)
   (b) Suppose, you have to select a site for new settlement among three sites. Construct a scoring matrix according to your preferable criteria (at least seven) for site selection. Mention which one is the best site and why. (20)
   (c) Briefly illustrate the basic street systems in a settlement? (07)
6. (a) "Sumerian civilization – large scale complex cities with well-defined social structure"- describe this with feudalism social structure. (10)
   (b) Show the cause -effect diagram of Egyptian civilization. (08)
   (c) Show evolution of Egyptian Pyramid in Egypt with necessary diagrams. (07)
   (d) What are the main features of Mesopotamian civilization? (10)
7. (a) Mention the name of the defined public space of Greek civilization. (12)
   "Their public assembly concept can be adopted for recent contemporary city planning"- give your opinion on behalf or against this statement considering a planner's view.
   (b) Differentiate among patricians, plebian and freedman. (08)
   (c) Define arch and vault concept. Show different types of roman arch techniques. (15)
8. Write short notes on the following (any five): (35)
   (a) Columns of Greek civilization (b) Ziggaraut pyramid (c) Set back rules
   (d) Parthenon (e) Grid iron pattern (f) Baroque city concept
   (g) Colosseum (h) Types of settlement pattern